B. T-primes

time limit per test 2 seconds
memory limit per test 256 megabytes
input stdin
output stdout

Rating: 1300 | Tags: binary search, implementation, math, number theory | https://codeforces.com/problemset/problem/230/B

We know that prime numbers are positive integers that have exactly two distinct positive divisors. Similarly, we'll call a positive integer *t* T-prime, if *t* has exactly three distinct positive divisors.

You are given an array of *n* positive integers. For each of them determine whether it is T-prime or not.

Input

The first line contains a single positive integer, *n* ($1 \le n \le 10^5$), showing how many numbers are in the array. The next line contains *n* space-separated integers x_i ($1 \le x_i \le 10^{12}$).

Please, do not use the `%lld` specifier to read or write 64-bit integers in C++. It is advised to use the `cin`, `cout` streams or the `%I64d` specifier.

Output

Print *n* lines: the *i*-th line should contain "YES" (without the quotes), if number x_i is T-prime, and "NO" (without the quotes), if it isn't.

Example

Input
3 4 5 6

Output
YES NO NO

Note

The given test has three numbers. The first number 4 has exactly three divisors — 1, 2 and 4, thus the answer for this number is "YES". The second number 5 has two divisors (1 and 5), and the third number 6 has four divisors (1, 2, 3, 6), hence the answer for them is "NO".